GWALIOR MUNICIPAL CORPORATION

WATER QUALITY SURVEILLANCE & ENVIRONMENTAL HEALTH CELL - ISO/IEC 17025:201

Central Analytical Testing Laboratory, Maharaj Bada Central Facility, Gwalior (M.P.)

NABL CERTIFIED TC-8491
Statutory BIS IS 10500:2012

POTABLE WATER QUALITY SPECTROSCOPIC BATCH ASSAY

Comprehensive statutory chemical, physical, and microbial assay under BIS IS 10500:2012

REPORT / TRACKING ID:	GWL-LAB-35698	**SAMPLING DATE:**	05 November 2026
MONITORING STATION:	Ward 14 - Maharaj Bada Central Plaza Hand Pump #04	**ANALYSIS PROTOCOL:**	Spectrophotometric Batch
ACCREDITED CHEMIST:	Dr. S. Mukherjee (Senior Analytical Chemist, TC-8491)	**SECTOR SEAL:**	SSE-KMS AES-256 Encrypted
AWS S3 OBJECT KEY:	`reports/lab-reports/gmc-ward-14/ward14_spectroscopic_assay_2026.pdf`		

#	WATER QUALITY PARAMETER	METHOD / INSTRUMENT	BIS IS 10500 LIMIT	RESULT	STATUS
01	**Potential of Hydrogen (pH)**	Electrometric Digital Glass Probe	6.5 - 8.5	**7.42**	**PASS**
02	**Nephelometric Turbidity**	Ratio Turbidimeter (Nephelometric)	Max 1.0 NTU	**0.65 NTU**	**PASS**
03	**Total Dissolved Solids (TDS)**	Gravimetric Drying at 180 C	Max 500 mg/L	**420 mg/L**	**PASS**
04	**Fluoride (as F)**	SPADNS Spectrophotometry 570nm	Max 1.0 mg/L	**0.85 mg/L**	**PASS**
05	**Nitrate (as NO3)**	UV Spectrophotometric Screening	Max 45 mg/L	**18.0 mg/L**	**PASS**
06	**Free Residual Chlorine**	DPD Colorimetric Comparator	0.2 - 1.0 mg/L	**0.45 mg/L**	**PASS**
07	**Sulphate (as SO4)**	Turbidimetric Method (BaCl2)	Max 200 mg/L	**34.5 mg/L**	**PASS**
08	**Total Coliform Organisms**	Multiple Tube Fermentation (MPN)	0 CFU/100ml	**0 CFU/100ml**	**PASS**
09	**Escherichia Coli (E. Coli)**	Membrane Filtration Method	0 CFU/100ml	**0 CFU/100ml**	**PASS**

STATUTORY VERDICT & FINDINGS:

VERDICT: Potable Drinking Water Specification Approved under BIS IS 10500:2012.

All observed parameters fall strictly within permissible statutory limits. Safe for distribution.

EXAMINED & CERTIFIED BY:

Dr. S. Mukherjee (Senior Analytical Chemist, TC-8491)

NABL National Accreditation Board Certification #TC-8491

GMC PHED SEAL
VERIFIED & SEALED
SHA-256: e3b0c442...

STATUTORY CLEARANCE AUTHORITY:

GMC Water Works Directorate, Gwalior

Public Health Engineering Department, M.P. Government